Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/10/2005
PI Name:	Duncan, Robert V. Ph.D.		
Project Title:	Critical Dynamics in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Albuquerque	**State:**	NM
Zip Code:	87131-1156	**Congressional District:**	1
Comments:	Web sites are for the PI (Duncan) and for the project (DYNAMX)		
Project Type:	Ground	**Solicitation / Funding Source:**	OTHER
Start Date:	10/01/1993	**End Date:**	03/31/2006
No. of Post Docs:	3	**No. of PhD Degrees:**	4
No. of PhD Candidates:	1	**No. of Master' Degrees:**	5
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	12
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Note - strayer says changed to ground in FY 2004.		
Key Personnel Changes/Previous PI:	Duncan, Robert		
COI Name (Institution):	Goodstein, David (California Institute of Technology) Liu, Feng-Chuan (Jet Propulsion Laboratory) Israelsson, Ulf (Jet Propulsion Laboratory) Chui, Talso (Jet Propulsion Laboratory)		
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

Task Description:	The nature of continuous (sometimes called 'second-order') transitions in systems driven well away from equilibrium is of fundamental scientific importance, yet not well understood. DYNAMX shall investigate such a system through measurements of the thermal gradients near the superfluid - normal fluid (He- II/He-I) interface while it is subjected to a heat flux Q in a microgravity environment. These thermal profile measurements shall be used to study the effective thermal conductivity very close to the interface where linear response has been observed to fail. Microgravity conditions are required in these measurements to greatly reduce static pressure variations across the sample under study, thus providing a more homogeneous sample in which to study dynamic critical phenomena. Results obtained on orbit are predicted to be intrinsically different from measurements on earth. This is true since the hydrostatic pressure gradient across the sample on earth prevents the correlation length from diverging. This fundamentally changes the nature of this continuous phase transition. Hence the real nature of this continuous phase transition to long range quantum order may only be observed (for the first time) on earth orbit. This project was deleted from the Flight Program in 2004, and is now in the ground program seeking a new flight opportunity. Earths gravity stabilizes the He-I/He-II interface in ground-based research. However, in microgravity only very precise heat flux control may be used to stabilize this interface. Hence, two major technical advances were required to stabilize and accurately measure these thermal properties in microgravity. First, thermometers have recently been developed with adequate temperature resolution to measure these very small temperature changes (as small as one tenth of one billionth of a degree) without creating perturbations with its self-absorbed heat from the charged particle radiation on orbit. Secondly, the experimental cell and support structure must provide an extremely high level of thermal isolation between thermometry probes along the column of helium, and must possess adequate stray heat control from other stages within the cryoprobe. In order to obtain accurate thermal gradient information close to the interface, and hence very close to criticality, corrections due to imperfections within the experimental cell must be carefully measured and controlled. These innovations have now been achieved at the level necessary to meet our exceptionally demanding science requirements. We intend to make measurements of the interface thermal profiles using heat flux values Q over the range 5 nW/cm2 < Q < 200 nW/cm2 on earth orbit. The superfluid transition represents a transition to a state of long-range quantum order. Such transitions under non-equilibrium conditions are interesting in the study of non-equilibrium superconductivity, shear transitions in some polymers, superconductors within a magnetic field, neutron stars, and theoretical studies in cosmology. These studies may become relevent to microelectronic miniaturization when line widths and line spacings approach atomic dimensions, which could occur as early as 2015 at the current rate of technology advancement.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Extensive ground based data have been obtained over the life of this project, and these have been published in refereed journals with international circulation, such as Physical Review Letters. Many leaders within science and technology in the USA have earned their degrees within this program, including staff physicists at national laboratories, and the Academy Professor of Physics at West Point. Activities, such as ours, advance our knowledge of fundamental physics while assuring that the United States remains at the forefront of basic research throughout the world.
Task Progress:	Our experiment has recently been down-graded from the flight program, so much of our effort has centered on the archival documentation of our 117 released flight drawings, our procedures, and other information in support of our flight build effort at UNM. We have also worked with our staff to arrange smooth employment transitions, and we are on target to achieve the necessary staff reductions that are mandated by this change in flight status. The one remaining Ph.D. student on this effort is scheduled to complete his Ph.D. in October, 2005. We are currently in the ground program, with science funding through half of FY 2006. We have met with JPL, and they have agreed to maintain the in-process flight hardware for the facility that was to support our experiments in flight for three years. This facility, called the Low Temperature Microgravity Facility (LTMPF), was delivered to JPL from Ball Aerospace, and it is approximately 75% complete. We have received one set of flight prototype electronics from Design Net Engineering at UNM, and we have made arrangements to continue these items, and the flight hardware that we have built as part of our science instrument, in secure storage. We hope to attract a future flight opportunity. This has been a remarkably productive year scientifically, as displayed in the Bibliography section of the Taskbook. In addition to these articles in print, we plan to publish four peer-reviewed papers in Physica, and we will make five presentations at the 24th International Conference on Low Temperature Physics (LT-24), two collaboratively with the CQ Experiment of Caltech (PI David Goodstein). Professor Raymond Nelson, Academy Professor of Physics at West Point, visited UNM during the Summer of 2004 and worked with our research group on the development of new PdMn thin films for bolometry and thermometry, and this work is now accepted for publication. 6.6-micron-thick films and screens were made by sputtering 0.4 atomic percent Mn in balance Pd onto sapphire and glass substrates. These structures were fabricated by Professor Nelson at the United States Military Academy through their local contractors, using a procedure similar to the one developed by Prof. Nelson while he was a Ph.D. student at UNM. The paramagnetic susceptibility of these films were measured as a function of temperature and magnetic field using a SQUID within a vacuum cryostat at UNM. These measurements will be published in Physica as part of the Proceedings of LT-24. A new method of stabilizing the temperature of a liquid helium cell to unprecedented precision was developed at UNM and published in the Journal of Low Temperature Physics in January, 2005, as part of the Proceedings of the 2004 International Quantum Fluids and Solids Symposium that was held in Trento, Italy. PI Duncan gave a talk on the development of this new technology at QFS 2004. This system uses the PdMn thermometric material to construct the cell sidewall, so the helium within the cell is in excellent thermal contact with the PdMn, thus greatly reducing the noise that results from energy fluctuations within the link between the helium and the Mn spin system. A series array of Josephson junctions, biased by millimeter waves at 94.000000000 GHz, are used to precisely control a heater mounted on this cell. This permits the power dissipated on the cell to be controlled at an accuracy approaching one part in a trillion. The power dissipated on this cell was changed by changing the quantum number of the rf-locked Josephson array. Once in weightless conditions this new system should be able to maintain a fixed-point temperature to within 50 pK near 2.2 K for an indefinitely long period of time. This technology may prove useful as an ultra-stable blackbody reference, one that is even more stable than the anticipated cooling rate of the Cosmic Microwave Background temperature, which is predicted to cool at about 200 pK per year. PI Duncan has been on sabbatical at Caltech throughout this year as a Gordon and Betty Moore Distinguished Scholar. He re-located one major cryostat from UNM to Caltech, and this apparatus has been used to develop superior new methods of experimental control, using the 'T4.5' flight prototype hardware. These new methods include an optimal PI/P controller, a Kalman filter, and a median filter to remain robust against charged particle strikes on Earth orbit. This apparatus will also be used to obtain more scientific data in a collaborative effort with the CQ experiment, which as a Guest Experiment to Critical Dynamics in

	Microgravity prior to the cancellation of this program. Duncan and the DX group collaborated closely with the CQ Experiment group at Caltech, and this collaboration has lead to the first measurement of the heat capacity of the SOC state. These important and difficult measurements were lead by Mr. Andrew Chatto, and they will be the centerpiece of his Ph.D. dissertation. PI Duncan served as the guest editor for a special edition of Advances in Space Research on Fundamental Physics Experiments in Space, and this came into print in June, 2005. Duncan has also worked with three other prior NASA flight PIs (lead by Dr. Martin Barmatz of JPL, PI of the MISTE experiment, and with Professor John Lipa of Stanford, PI on the SUMO Experiment and Dr. Inseob Hahn of JPL, a Guest Investigator on MISTE) to co-author a major review article on Critical Phenomena Measurements in Space, which is currently under editorial review and tentatively scheduled for publication in the January, 2006 edition of Reviews of Modern Physics. Duncan conducted his weekly group meetings at UNM on Mondays, and spent the remainder of his week at Caltech on sabbatical. Two invited scientific talks have been delivered on this research during the last year, including the Caltech Physics Research Symposium by PI Duncan in October, 2004, and the Caltech Condensed Matter Physics Seminar by Dr. Steve Boyd in November, 2004. Dr. Boyd also delivered two contributed talks to the March Meeting of the American Physical Society in 2005. A new method of using synchronous demodulation to measure the speed and the attenuation of the new temperature / entropy wave that propagates only against the vector direction of the heat flux on the SOC state has been developed at UNM. This technique is being used now to obtain more data on both the ‘T4d’ cryostat at UNM, and within Andrew Chatto’s cryostat at Caltech. This new technique will be reported on at LT-24, along with new data on this wave’s speed and attenuation as a function of the heat flux Q that was used to self-organize the liquid helium column. Dr. Steve Boyd has recently observed that in the two-fluid model of superfluid hydrodynamics without dissipation a thermal gradient results under Earth’s gravity that does not reverse when the direction of the heat flux through the helium is reversed. This surprising result will be reported by Dr. Boyd in a talk at LT-24. Two major flight prototype experiments are currently cold and collecting data. The 'T4d' prototype is operational at UNM, and being used to measure the velocity and attenuation of a new temperature / entropy wave that we discovered experimentally last year that travels only against the heat flux direction on the self-organized critical (SOC) state. The other apparatus, called the 'T4.5' flight prototype, is cold at Caltech as mentioned above, and currently being used to develop superior temperature control. This work has been published in the Proceedings of the American Controls Conference 2005, where graduate student Jinyang Liu was invited to present this work as a full paper. The 'T4.5' apparatus will soon be used to take collaborative data on dynamical enhancements of the heat capacity of liquid helium with the CQ experiment effort. Please direct all questions or requests for preprints, reprints, or further information to PI Robert Duncan at duncan@unm.edu.
Bibliography Type:	Description: (Last Updated: 12/07/2006)
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